

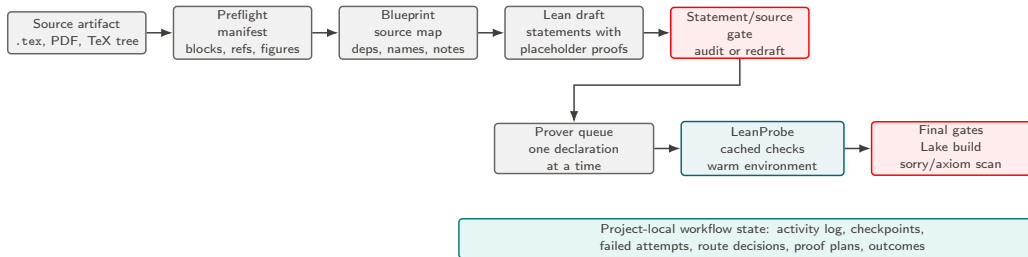
EPFLemma Autoformalization Agent

Fully automated formalization of two mathematical papers

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Case studies: *Parametrization of Pythagorean Triples by a Single Triple of Polynomials* and *A Calculus Proof of the Cramer–Wold Theorem*.

Full Workflow: Document to Verified Lean Project



Idea. The model edits mathematics, but the runtime controls the project: source resolution, review gates, theorem assignment, verification, and resumability.

Work organization. Blueprinting creates the task list; statement review fixes the target; the prover queue assigns one theorem at a time.

Zoom In: Formalization Before Proof Search

Pre-proof contract

1. Resolve the source and pinned Lean environment.
2. Extract theorem-like blocks, labels, references, PDFs, and support files.
3. Build a blueprint mapping source spans to Lean declarations.
4. Draft declarations with placeholder theorem proofs.
5. Review statement faithfulness before proving.

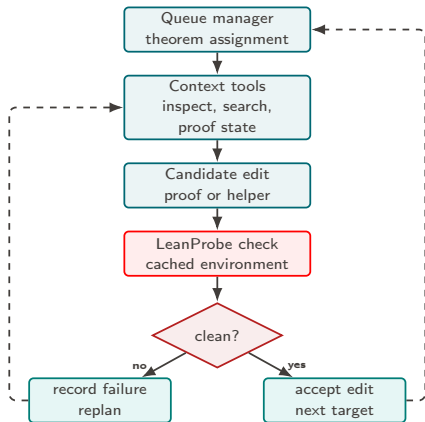
The important separation: **type-checking a statement skeleton** is not the same as checking that it says the paper's theorem.

What the review rejects

Check	Typical failure
Statement type	Proves a different claim
Quantifiers	Dependency/order changed
Variable types	Wrong domain/codomain
Hypotheses	Added or missing side condition
Conclusion	Weakened equality/existence
Encoding bridge	Missing recovery lemmas
Hygiene	Hidden axiom, admit, unsafe, sorry

In the Pythagorean case, this prevented a buildable but weaker function-level translation of an integer-valued polynomial theorem.

Queue-Managed Proof Workflow



Why the queue matters

- The model sees the active theorem, local context, and failed attempts.
- It cannot advance unless the verifier accepts the assigned theorem.
- Failed proof shapes become theorem-local memory.
- LeanProbe gives fast candidate checks; Lake is the final gate.

Pipeline Tested on Two Full Papers

What counts as success here

Starting from the paper source, EPF_{LEMMA} builds a reviewed Lean skeleton, completes the proof obligations, and finishes with a Lake build and no unapproved sorry/axiom leftovers.

1. Pythagorean triples

Parametrization of Pythagorean Triples by a Single Triple of Polynomials

- Area: number theory and integer-valued polynomials.
- Main issue: the witnesses must be polynomial objects, not just functions evaluated on integer inputs.

2. Cramer–Wold theorem

A Calculus Proof of the Cramer–Wold Theorem

- Area: probability, measure theory, and analysis.
- Main issue: the paper uses a missing distribution-theory inversion fact as background.

The next two slides explain what each theorem says and why each one stresses a different part of the autoformalization pipeline.

Problem 1: Pythagorean Polynomial Parametrization

Area. Number theory, commutative algebra, integer-valued polynomials.

What the theorem says. Every integer solution of $x^2 + y^2 = z^2$ is produced by one explicit triple of rational polynomials which nevertheless takes integer values on all integer inputs.

$$T(a, b, c) = \left(\frac{c(a^2 - b^2)}{2}, cab, \frac{c(a^2 + b^2)}{2} \right)$$

Proof story.

- Use the classical Pythagorean parametrization $T(a, b, c)$.
- Show $T(a, b, c)$ is integral exactly when c is even or a, b have the same parity.
- Parametrize that parity condition by $a = y + zw$, $b = z - yw$, $c = 2x - xw$.
- Substitution gives the four-variable integer-valued polynomial triple.
- A separate obstruction says this cannot be done over $\mathbb{Z}[x]$ with one integer-coefficient triple.

Lean/formalization point

The key modeling choice is not just proving a formula after evaluation. The witnesses must be actual elements of

$$\text{Int}(\mathbb{Z}^n) \subseteq \mathbb{Q}[x_1, \dots, x_n],$$

so the formalization has both:

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RatPoly n and IsIntValued : RatPoly n -> Prop.
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Files. Basic, SourceLemmas, Obstructions, IntegerValued, Positive, Explanatory.

Why this was a good EPFLemma test. A buildable draft can accidentally formalize functions $\mathbb{Z}^4 \rightarrow \mathbb{Q}$ instead of polynomial objects. The statement/source gate catches that.

Problem 2: Cramer–Wold and Missing Analysis Infrastructure

Area. Probability and measure theory, integral geometry, PDE / distribution theory.

What the theorem says. A probability measure on $\mathbb{R}^n$ is completely determined by how much mass it assigns to every closed half-space.

$$\mu(S) = \nu(S) \text{ for all closed half-spaces } S \implies \mu = \nu$$

Proof story.

- Half-space values determine the average-distance function $f_\mu(y) = \int (\|y - x\| - \|x\|) d\mu(x)$ via a Crofton/Fubini argument.
- In odd dimension, f_μ determines μ : apply enough Laplacians to the distance kernel.
- Even dimensions reduce to odd dimensions by embedding $\mathbb{R}^{2m} \hookrightarrow \mathbb{R}^{2m+1}$.

Hidden background fact.

$$\Delta^{m+1} \|x\| = c \cdot \delta_0, \quad c \neq 0.$$

That Laplacian identity was not available as a ready-made Mathlib theorem. Closing the Cramer–Wold proof required building the missing analytic layer:

- tempered-distribution norm kernel;
- Fourier/Laplacian algebra for iterated Laplacians;
- PhysLean real-distribution bridge;
- transport between Euclidean-space models;
- final odd-dimensional inversion theorem.

Ablation Results on the Two Papers

Kimi-K2.6

Source	Cond.	Outcome	Calls	In tok.
Pyth.	Full	succ.	1043	46.9M
Pyth.	NoQ+Tools	fail	2000	160.2M
Pyth.	Q+CLI	succ.	1561	91.3M
Pyth.	NoQ+CLI	fail	2000	157.5M
C-W	Full	succ.	1278	66.0M
C-W	NoQ+Tools	fail	2000	127.8M
C-W	Q+CLI	fail	2000	113.1M
C-W	NoQ+CLI	fail	2000	167.0M

Takeaway for Kimi-K2.6: queue management changes completion behavior; no-queue variants hit the 2000-call budget on both document-level projects.

GPT-5.5

Source	Cond.	Outcome	Calls	In tok.
Pyth.	Full	succ.	258	14.3M
Pyth.	NoQ+Tools	succ.	568	45.0M
Pyth.	Q+CLI	succ.	277	14.3M
Pyth.	NoQ+CLI	succ.	984	89.9M
C-W	Full	succ.	495	25.9M
C-W	NoQ+Tools	succ.	448	32.8M
C-W	Q+CLI	succ.	959	59.5M
C-W	NoQ+CLI	succ.	476	49.7M

Takeaway for GPT-5.5: all variants finish, but full EPFL_{EMMA} usually lowers model-side token use and keeps the proof search more controlled.